

ABHISHEK ANIL

Biomedical Engineering Student | AI & Machine Learning

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Professional Summary

Biomedical Engineering undergraduate at M.S. Ramaiah Institute of Technology (CGPA: 8.06/10) with hands-on experience in Artificial Intelligence, Medical Imaging, Biomedical Signal Processing and Embedded Systems. Developed healthcare-focused deep learning solutions using TensorFlow and PyTorch, including respiratory sound classification (84% accuracy) and AI-assisted medical image segmentation. Passionate about building intelligent technologies that improve healthcare.

Education

M.S. Ramaiah Institute of Technology

B.E. Medical Electronics Engineering

Expected Graduation: 2027

CGPA: 8.06 / 10

Technical Skills

Programming: Python, C, C++, Java, MATLAB

AI & Machine Learning: TensorFlow, PyTorch, CNN, MobileNetV2, Computer Vision, OpenCV

Medical Imaging: MedSAM, TotalSegmentator, DICOM, MediaPipe, SimpleITK, pydicom, 3D Slicer

Development: React, Next.js, TypeScript, Git, GitHub, VS Code, Unity, Jupyter Notebook

Biomedical: Biomedical Signal Processing, Medical Electronics, Biomedical Instrumentation, Embedded Systems

Creative: Adobe Premiere Pro, After Effects, Illustrator, DaVinci Resolve, Canva

Projects

AI-Based Lung Sound Classification

- Developed a MobileNetV2-based deep learning pipeline using TensorFlow.
- Converted respiratory audio into Mel spectrograms for feature extraction.
- Achieved approximately 84% classification accuracy on lung sound datasets.

AI Medical Imaging Suite

- Built an AI-assisted workflow using MedSAM, TotalSegmentator, OpenCV and 3D Slicer.
- Implemented DICOM processing and automatic anatomical segmentation.
- Developed gesture-based image navigation using MediaPipe.

Midnight Engineering Portfolio

- **Technologies:** Next.js • React • TypeScript • Tailwind CSS • Framer Motion
 - Designed and developed a modern engineering portfolio with a modular, component-based architecture using Next.js and TypeScript.
 - Implemented smooth scrolling, interactive animations, dark/light theme support, reusable UI components, and responsive layouts to deliver a polished user experience.
 - Built a scalable project showcase architecture with dedicated case-study pages, enabling future engineering and AI projects to be presented in a structured, maintainable format.

Unity Fundamentals Project

- Developed an interactive Unity prototype featuring scene design, object movement, audio integration and introductory VR concepts.

Academic Projects

Embedded Biomedical Sensor Interface

- Interfaced biomedical sensors with a microcontroller for real-time data acquisition and monitoring.
 - Implemented sensor data acquisition, processing, and display for embedded healthcare applications.
 - Diagnosed and resolved hardware communication issues to improve system reliability and performance.
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Biomedical Analog Circuit Design

- Designed operational amplifier-based biomedical signal conditioning circuits.
- Performed frequency response, gain, and stability analysis for analog electronic systems.
- Developed a 555 timer-based **Fastest Fingers First** system for educational and event applications.

Student Finance Management System (C++)

- Developed a C++ application to manage personal income and expense records.
 - Implemented categorized financial tracking using modular programming concepts.
 - Applied debugging, data structures, and object-oriented programming principles to improve application reliability.
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Lifestyle Preference Analyzer (Java)

- Developed a Java application that analyzes lifestyle preferences using decision-based logic.
- Applied object-oriented programming concepts, multithreading, and concurrent execution techniques.
- Strengthened understanding of Java application architecture and software design principles.

Leadership & Activities

- Technical Fest Event Coordinator — Coordinated technical competitions, participant flow and event logistics.
- Senior Member — Studio RIT (Former Video Editing Head) — Mentored junior editors and contributed to creative media production.
- Collaborated on biomedical AI dataset preprocessing and model development.

Languages

English • Hindi • Malayalam • Japanese (Conversational)